

TASK II

Identification of Entities, Attributes, Primary Keys and Relationships

System: Flipkart E-Commerce Platform

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1. Introduction

This task identifies the main data required for a basic Flipkart e-commerce system. Each entity has a primary key, attributes and foreign keys used to connect related data.

2. Entities and Attributes

PK = Primary Key **FK** = Foreign Key

Entity	Primary Key	Important Attributes
Customer	customer_id	name, email, password_hash, phone
Address	address_id	customer_id (FK), address, city, state, postal_code
Seller	seller_id	business_name, email, GST_number, status
Category	category_id	category_name
Product	product_id	category_id (FK), name, brand, description
Product Listing	listing_id	product_id (FK), seller_id (FK), price, stock
Cart	cart_id	customer_id (FK), created_at
Cart Item	cart_id + listing_id	quantity, added_at
Wishlist	wishlist_id	customer_id (FK), created_at
Wishlist Item	wishlist_id + product_id	added_at
Customer Order	order_id	customer_id (FK), address_id (FK), date, status, total
Order Item	order_item_id	order_id (FK), listing_id (FK), quantity, price
Payment	payment_id	order_id (FK), method, amount, status
Delivery	delivery_id	order_id (FK), tracking_number, delivery_status
Review	review_id	customer_id (FK), product_id (FK), rating, comment

3. Main Relationships

- Customer 1 : M Address
- Customer 1 : 1 Cart
- Cart 1 : M Cart Item
- Product Listing 1 : M Cart Item
- Customer 1 : 1 Wishlist
- Wishlist 1 : M Wishlist Item
- Product 1 : M Wishlist Item
- Category 1 : M Product
- Product 1 : M Product Listing
- Seller 1 : M Product Listing
- Customer 1 : M Customer Order
- Customer Order 1 : M Order Item
- Product Listing 1 : M Order Item
- Customer Order 1 : M Payment
- Customer Order 1 : 1 Delivery
- Customer 1 : M Review
- Product 1 : M Review

4. Conclusion

The identified entities and relationships cover the basic functions of the e-commerce system. They can now be used to create the ER diagram in StarUML.